



Girraween High School
Mathematics (Extension 1)

Yr 12 – Task 3 (2007)
Time allowed – 90 minutes

DIRECTIONS TO CANDIDATES

- All necessary working should be shown.
- Marks may be deducted for careless or badly arranged work
- Start each question on a *new* sheet of paper.

Question 1 (17 marks)

Marks

- (a) Consider the function $y = 2 \sin^{-1}\left(\frac{x}{3}\right)$. ~~Sketch the function.~~
- (i) State the domain and range 3
- (ii) Evaluate $f(0)$ 1
- (iii) Draw the graph of $y = f(x)$ 3
- (b) Draw a sketch of the curve, $y = (x - 3)^2 - 3$.
- (i) Find the largest positive domain such that the graph defines a function $f(x)$ which has an inverse 2
- (ii) Find the inverse function, stating its domain and range. 4
- (iii) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ on the same axes. 4

Question 2 (11 marks)

- (a) Evaluate in terms of π .
- (i) $\cos^{-1}\left(-\frac{1}{2}\right) - \sin^{-1}\left(-\frac{1}{2}\right)$ 2
- (ii) $\cos^{-1}\left(\sin^2 \frac{\pi}{4}\right)$ 2
- (b) Evaluate as a rational number.
- $\tan\left(2 \tan^{-1}\left(-\frac{1}{2}\right)\right)$ 3

- | | |
|--|--------------|
| (b) Using $\sin(A+B) = \sin A \cos B + \cos A \sin B$, prove that | Marks |
| $\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13} = \sin^{-1} \frac{56}{65}$. | 4 |

Question 3 (17 marks)

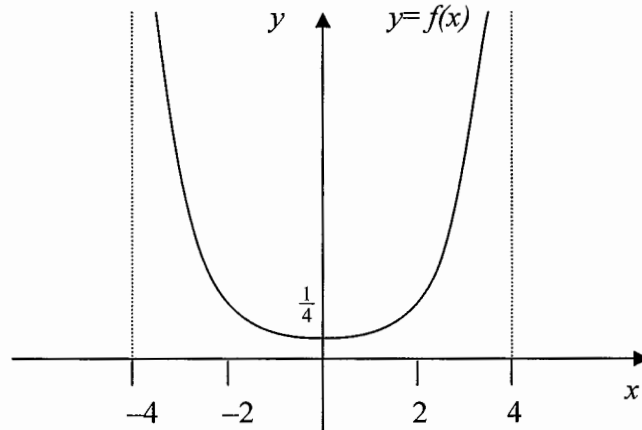
- | | |
|--|---|
| (a) Differentiate the following | |
| (i) $y = \cos^{-1} 2x$ | 2 |
| (ii) $y = \tan^{-1} \sqrt{x}$ | 3 |
| (iii) $y = \frac{1}{\sin^{-1} x}$ | 3 |
| (iv) $y = x \sin^{-1} x + \sqrt{1-x^2}$ | 4 |
| (b) Show that the curves $y = \cos^{-1} x$ and $y = 2 \tan^{-1}(1-x)$ intersect the y-axis at the same point, and have a common tangent at this point. | 5 |

Question 4 (17 marks)

- | | |
|---|---|
| (a) Find the following integrals. | |
| (i) $\int \frac{dx}{1+9x^2}$ | 3 |
| (ii) $\int \frac{5dx}{x^2+2}$ | 3 |
| (iii) $\int_0^{\frac{3}{2}} \frac{dx}{\sqrt{9-2x^2}}$ | 3 |
| (iv) $\int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}}$ | 3 |

(b)

Marks



Let $f(x) = \frac{1}{\sqrt{16-x^2}}$. The graph of $y = f(x)$ is sketched above.

- (i) Show that $f(x)$ is an even function. 2
- (ii) Find the area enclosed by $y = f(x)$, the x -axis, $x = 2$, and $x = -2$. 3

Question 5 (19 marks)

(a) Find the following integrals using the given substitution.

(i) $\int x\sqrt{x^2+2}dx$ $u = x^2 + 2$ 3

(ii) $\int \frac{x^2 dx}{\sqrt{1-x^3}}$ $u = 1 - x^3$ 4

(iii) $\int x\sqrt{x+1}dx$ $x = u^2 - 1$ 4

(b) Evaluate following integrals using the given substitution.

(i) $\int_0^1 x(1+x^2)^2 dx$ $u = x^2 + 1$ 4

(ii) $\int_0^{\frac{\pi}{6}} \frac{2\cos x dx}{1+4\sin^2 x}$ $u = 2\sin x$ 4

Question 6 (14 marks)**Marks**(a) Find the general solution (in terms of π) of the following functions.

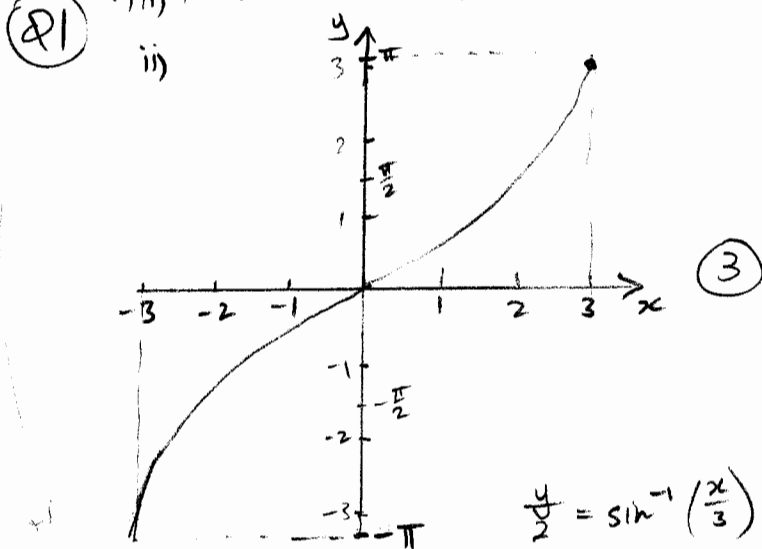
$$(i) \sin \theta = \frac{1}{2} \quad 3$$

$$(ii) \cos \theta = -\frac{1}{\sqrt{2}} \quad 3$$

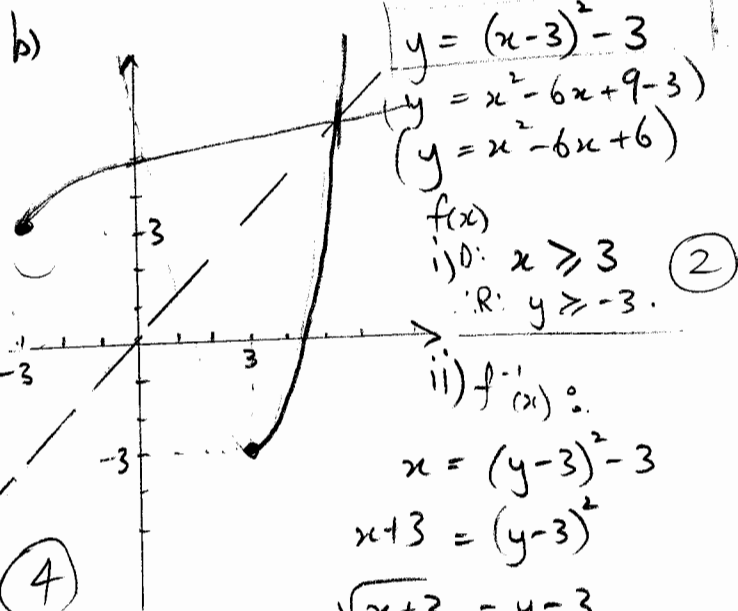
(b) Given that, $x^2 + 4x + 5 = (x + a)^2 + b$ (i) Find a and b . 2(ii) Hence evaluate $\int \frac{dx}{x^2 + 4x + 5}$ 2(c) The function $f(x) = \sec x$ for $0 \leq x < \frac{\pi}{2}$, and is not defined for other values of x .(i) State the domain and range of the inverse function $f^{-1}(x)$ 1(ii) Show that, $f^{-1}(x) = \cos^{-1}\left(\frac{1}{x}\right)$. 1(iii) Hence find, $\frac{d}{dx} f^{-1}(x)$. 2

TASK 3 (2007) SOLUTIONS

Q1 a) i) $f(0) = 2\sin^{-1}(\frac{0}{3}) = 0$ (1)



(i) $D: -1 \leq \frac{x}{3} \leq 1$ $R: -\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$ (3)
 $D: -3 \leq x \leq 3$ $R: -\pi \leq y \leq \pi$



$f^{-1}(x): y = \sqrt{x+3} + 3$

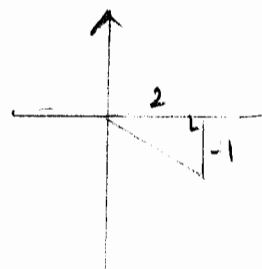
$D: x \geq -3$ (4)

$R: y \geq 3$

Q2 a) i) $\cos^{-1}(-\frac{1}{2}) - \sin^{-1}(-\frac{1}{2})$
 $= \frac{2\pi}{3} - (-\frac{\pi}{6})$
 $= \frac{5\pi}{6}$ (2)

ii) $\cos^{-1}(\sin^2 \frac{\pi}{4})$
 $= \cos^{-1}((\frac{1}{\sqrt{2}})^2)$
 $= \cos^{-1}(\frac{1}{2})$
 $= \frac{\pi}{3}$ (2)

iii) $\tan(2 \tan^{-1}(-\frac{1}{2}))$
 $= \tan(2\theta)$
 $= \frac{2 \tan \theta}{1 - \tan^2 \theta}$
 $= \frac{2(-\frac{1}{2})}{1 - (-\frac{1}{2})^2}$
 $= \frac{-1}{1 - \frac{1}{4}}$
 $= -\frac{4}{3}$ (3)

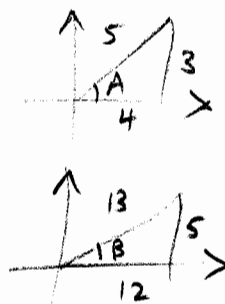


b) If $\sin^{-1} \frac{3}{5} + \sin^{-1} \frac{5}{13}$
 $(A) + (B)$

$\sin(A+B) = \sin A \cos B + \cos A \sin B$
 $= \frac{3}{5} \cdot \frac{12}{13} + \frac{4}{5} \cdot \frac{5}{13}$
 $= \frac{36}{65} + \frac{20}{65}$

$\sin(A+B) = \frac{56}{65}$ (4)

$\therefore A+B = \sin^{-1} \frac{56}{65}$



Q3 a) i) $y = \cos^{-1} 2x$

$$y' = \frac{-1}{\sqrt{1-(2x)^2}} \times 2$$

$$y' = \frac{-2}{\sqrt{1-4x^2}}$$

(2)

ii) $y = \tan^{-1} \sqrt{x}$

$$y' = \frac{1}{1+(\sqrt{x})^2} \times \frac{1}{2} x^{-\frac{1}{2}}$$

$$= \frac{1}{1+x} \times \frac{1}{2\sqrt{x}}$$

$$y' = \frac{1}{2\sqrt{x}(1+x)}$$

(3)

iii) $y = \frac{1}{\sin^{-1} x}$

$$y = (\sin^{-1} x)^{-1}$$

$$y' = -(\sin^{-1} x)^{-2} \times \frac{1}{\sqrt{1-x^2}}$$

$$y' = \frac{-1}{(\sin^{-1} x)^2 \sqrt{1-x^2}}$$

(3)

iv) $y = x \sin^{-1} x + \sqrt{1-x^2}$

$$= x \cdot \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x \cdot 1 + \frac{1}{2} (1-x^2)^{-\frac{1}{2}} \times -2x$$

$$= \frac{x}{\sqrt{1-x^2}} + \sin^{-1} x - \frac{x}{\sqrt{1-x^2}}$$

$$= \sin^{-1} x$$

(4)

b) $y = \cos^{-1} x$ and $y = 2 \tan^{-1}(1-x)$

when $x=0$ $y = \cos^{-1}(0)$
 $y = \frac{\pi}{2}$

$y = 2 \tan^{-1}(1-0)$
 $y = 2 \left(\frac{\pi}{4} \right)$
 $y = \frac{\pi}{2}$

$\therefore$ Both intersect the y-axis at the point $(0, \frac{\pi}{2})$

$$y' = \frac{-1}{\sqrt{1-x^2}}$$

$$y' = \frac{2}{1+(1-x)^2} x^{-1}$$

when $x=0$ $y' = \frac{-1}{\sqrt{1-(0)}}$

$$y' = -1$$

$$y' = \frac{-2}{1+(1-0)^2}$$

$$y' = -1$$

$\therefore$ Both curves have common tangent at the point $(0, \frac{\pi}{2})$

(5)

$$\begin{aligned}
 \textcircled{Q4} \text{ a) i) } & \int \frac{dx}{1+9x^2} \\
 &= \frac{1}{9} \int \frac{1}{\left(\frac{1}{3}\right)^2 + x^2} dx \\
 &= \frac{1}{9} \times 3 \tan^{-1} 3x + C \\
 &= \frac{1}{3} \tan^{-1}(3x) + C \quad \textcircled{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{ii) } & \int \frac{5}{x^2+2} dx \\
 &= 5 \int \frac{1}{(\sqrt{2})^2 + x^2} dx \\
 &= 5 \times \frac{1}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C \\
 &= \frac{5}{\sqrt{2}} \tan^{-1}\left(\frac{x}{\sqrt{2}}\right) + C \quad \textcircled{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{iii) } & \int_0^{3/2} \frac{dx}{\sqrt{9-2x^2}} \\
 &= \frac{1}{\sqrt{2}} \int_0^{3/2} \frac{1}{\sqrt{\frac{9}{2}-x^2}} dx \\
 &= \frac{1}{\sqrt{2}} \left[\sin^{-1}\left(\frac{\sqrt{2}x}{3}\right) \right]_0^{3/2} \\
 &= \frac{1}{\sqrt{2}} \left(\sin^{-1}\left(\frac{\sqrt{2}}{2}\right) - \sin^{-1}(0) \right) \\
 &= \frac{1}{\sqrt{2}} \left(\frac{\pi}{4} \right) \\
 &= \frac{\pi}{4\sqrt{2}} \quad \textcircled{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{iv) } & \int_1^{\sqrt{3}} \frac{dx}{\sqrt{4-x^2}} \\
 &= \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_1^{\sqrt{3}} \\
 &= \sin^{-1} \frac{\sqrt{3}}{2} - \sin^{-1} \frac{1}{2} \\
 &= \frac{\pi}{3} - \frac{\pi}{6} \\
 &= \frac{\pi}{6} \quad \textcircled{3}
 \end{aligned}$$

$$\text{b) i) } f(x) = \frac{1}{\sqrt{16-x^2}}$$

$$f(-x) = \frac{1}{\sqrt{16-(-x)^2}} = \frac{1}{\sqrt{16-x^2}}$$

$$\therefore f(x) = f(-x) \therefore \text{an EVEN FUNCTION} \quad \textcircled{2}$$

$$\begin{aligned}
 \text{ii) } A &= 2 \times \int_0^2 \frac{1}{\sqrt{16-x^2}} dx \\
 &= 2 \times \left[\sin^{-1}\left(\frac{x}{4}\right) \right]_0^2 \\
 &= 2 \times \left(\sin^{-1} \frac{1}{2} - \sin^{-1} 0 \right) \\
 &= 2 \left(\frac{\pi}{6} - 0 \right) \\
 &= \frac{\pi}{3} \quad \textcircled{3}
 \end{aligned}$$

Q5 a) i) $\int x \sqrt{x^2+2} dx$ $u = x^2+2$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$
 $= \frac{1}{2} \int \sqrt{u} du$
 $= \frac{1}{2} \times \frac{2}{3} u^{3/2} + C$
 $= \frac{1}{3} (x^2+2)^{3/2} + C \left(= \frac{\sqrt{(x^2+2)^3}}{3} + C \right)$ (3)

ii) $\int \frac{x^2}{\sqrt{1-x^3}} dx$ $u = 1-x^3$
 $du = -3x^2 dx$
 $-\frac{1}{3} du = x^2 dx$
 $= -\frac{1}{3} \int \frac{du}{\sqrt{u}}$
 $= -\frac{1}{3} \int u^{-1/2} du$
 $= -\frac{1}{3} \times 2u^{1/2} + C$
 $= -\frac{2}{3} (1-x^3)^{1/2} + C \left(= -\frac{2\sqrt{1-x^3}}{3} + C \right)$ (4)

iii) $\int x \sqrt{x+1} dx$ $x = u^2-1$
 $du = 2u du$
 $= \int (u^2-1) u \cdot 2u du$
 $= \int 2u^4 - 2u^2 du$ also $u = \sqrt{x+1}$
 $= \frac{2u^5}{5} - \frac{2u^3}{3} + C$
 $= \frac{2}{5} (\sqrt{x+1})^5 - \frac{2}{3} (\sqrt{x+1})^3 + C$ (4)

b) i) $\int_0^1 x (1+x^2)^2 dx$ $u = x^2+1$
 $x=1, u=2$
 $x=0, u=1$
 $du = 2x dx$
 $\frac{1}{2} du = x dx$
 $= \frac{1}{2} \int_1^2 u^2 du$
 $= \frac{1}{2} \left[\frac{u^3}{3} \right]_1^2$
 $= \frac{1}{2} \left(\frac{8}{3} - \frac{1}{3} \right)$
 $= \frac{7}{6}$ (4)

ii) $\int_0^{\pi/6} \frac{2 \cos x}{1+4 \sin^2 x} dx$ $u = 2 \sin x$
 $du = 2 \cos x dx$
 $= \int_0^1 \frac{du}{1+u^2}$ $x = \pi/6, u = 1$
 $x = 0, u = 0$
 $= [\tan^{-1} u]_0^1$
 $= \tan^{-1}(1) - \tan^{-1}(0)$
 $= \frac{\pi}{4} - 0$
 $= \pi/4$ (4)

(Q6) c) $f(x) = \sec x$ $D: 0 \leq x < \frac{\pi}{2}$
 $\therefore R: y \geq 1$

i) $\therefore f^{-1}(x)$ $D: x \geq 1$
 $R: 0 \leq y < \frac{\pi}{2}$ (1)

ii) $f^{-1}(x): x = \sec y$
 $x = \frac{1}{\cos y}$
 $\cos y = \frac{1}{x}$
 $f^{-1}(x): y = \cos^{-1}\left(\frac{1}{x}\right)$ (1)

iii) $\frac{d}{dx} f^{-1}(x)$
 $= \frac{-1}{\sqrt{1 - \left(\frac{1}{x}\right)^2}} \times -1 x^{-2}$
 $= \frac{1}{x^2 \sqrt{\frac{x^2 - 1}{x^2}}}$ (2)
 $= \frac{1}{x \sqrt{x^2 - 1}}$

b) i) $x^2 + 4x + 5 = x^2 + 4x + 4 + 1$
 $= (x+2)^2 + 1$
 $\therefore a=2, b=1$ (2)

ii) $\int \frac{1}{x^2 + 4x + 5} dx$
 $= \int \frac{1}{1 + (x+2)^2} dx$
 $= \tan^{-1}(x+2) + C$ (2)

a) i) $\sin \theta = \frac{1}{2}$

$\theta = \sin^{-1}\left(\frac{1}{2}\right) + 2n\pi$ or $(\pi - \sin^{-1}\left(\frac{1}{2}\right)) + 2n\pi$

$\theta = \frac{\pi}{6} + 2n\pi$; $\frac{5\pi}{6} + 2n\pi$

where n is an integer. (3)

ii) $\cos \theta = -\frac{1}{\sqrt{2}}$

$\theta = 2n\pi \pm \cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

$\theta = 2n\pi \pm \frac{3\pi}{4}$ where n is an integer. (3)